

Lesson 1

Equipment: Scratch code editor, provided sheets (including paper with printed shapes), pens/pencils

Summary

- This lesson will allow pupils to explore Scratch's pen tool. Pupils should become familiar with basic Scratch commands. Experience using these Scratch commands should help pupils when they use similar commands with Python's turtle module in later lessons. Pupils will be encouraged to consider drawing simple shapes and how they could describe this process to others.

Learning Intentions

- I can recognise that programming is giving instructions to a computer.
- I can use the following Scratch blocks to create simple programs: pendown, penup, forward, left and right.
- I can devise a Scratch program that draws simple shape(s).
- I can visualise, without the use of a computer, the execution of a Scratch program.

Curriculum Links

- TCH 2-13a: pupils create programs using their instructions for how to draw a shape. This involves pupils considering the process they follow to draw the shape and using this to create a program which follows these steps. This is intended to encourage pupils to use computational thinking to model the process of drawing.
- TCH 2-14a: pupils should be comfortable using and explaining the following Scratch blocks: penup, pendown, forward, left and right.
- TCH 2-15a: pupils will create their own short programs that draw a given shape.

Plan

Starter (5 minutes)

- Prompt for pupils: Think throughout this lesson about how we give instructions. How is giving instructions to a computer different/similar to giving instructions to other people?

Unplugged (20 minutes)

Part 1

- Instruct pupils to get into pairs
- Distribute to all pupils a small piece of paper with a shape printed on.
- Instruct pupils to write instructions to verbally give their partner that describe how to draw their shape.
- Tell pupils there is an additional challenge to this as their partner will have their eyes closed while following their instructions.

Part 2

- Pupils take turns giving instructions to each other. The person drawing should have their eyes closed.

Part 3

- As a class, pupils will reflect on their experiences of this activity. Pupils should be encouraged to reflect on instructions that were helpful vs. not helpful.

Taught Section (5-10 minutes)

- Demonstrate the use of Scratch's pen tool. A suggested method for this is to draw a square while getting pupils to predict the next instruction at each step as this uses VPC.
- Ensure pupils have been shown how to use penup, pendown, forward, left and right. It is a good idea to show pupils the erase all block so they can easily clear their screens.

Coding Activity (30 minutes)

- In pairs, pupils choose one of the shapes they drew during the unplugged activity.
- Pupils have 10 minutes to write Scratch code **on paper** to draw their shape.
- Pupils are then given 20 minutes to write their Scratch code on the code editor.
- Pupils should note any changes that had to be made to their original code on their reflection sheets.

Closing (10 minutes)

- Pupils are encouraged to consider similarities and differences between giving instructions to their friend compared to writing instructions using Scratch.
- Encourage pupils to think about the potential for other people to misinterpret their meaning compared to the computer that always does exactly what they tell it (though not always what they want!).

Spatial Skills

- Use VPC during the taught section by showing pupils sections of Scratch code and getting them to predict what will happen when the code is run.
- Pupils will follow a VPC process to create their Scratch shape program. Ensure pupils follow the step of writing their code on paper **before** writing it on the Scratch editor to ensure they are visualising the execution of code as much as possible.

General advice for this lesson

- Use the provided Scratch pen cheatsheets to help pupils with writing their code on paper. Encourage pupils to work in their pairs to solve problems and to look at their cheatsheets if they get stuck.